

Block 0 — Set up your tools

Goal: install the tools for this course, get the course material with **Git**, and serve your first page with **Node**. At the end, you get Block 1 with Git — live, in class.

Time: ~45 minutes · **You need:** your laptop (or a lab PC) and internet

You don't have the course material on your computer yet. Read this sheet on GitHub until task 2c. After that, you can open it in VS Code.

The terminal: all commands on this sheet run in a terminal. Use the one in VS Code: **Terminal** → **New Terminal** (Ctrl + `).

⚠ After you install a tool, **close VS Code completely and open it again**. A terminal that was already open doesn't know the new command yet.

Task 1 — VS Code and Chrome

The lab PCs have both. On your own laptop, install what's missing:

- **VS Code:** <https://code.visualstudio.com>
- **Chrome:** <https://www.google.com/chrome>

Task 2 — Git

2a — Install Git

- **Windows:** download the installer from <https://git-scm.com/downloads/win>. Keep the default settings.
- **macOS:** run `git --version` in a terminal. If Git is missing, macOS offers to install the *Command Line Developer Tools*. Accept, and wait until it's done.
- **Linux:** install it with your package manager, e.g. `sudo apt install git`.

Check it — this must print a version number:

```
git --version
```

2b — Set up Git

Git writes your **name** and **email** into every commit you make:

```
git config --global user.name "Your Name"
git config --global user.email "you@example.com"
```

Sometimes Git opens an editor, e.g. for the message of a merge. By default, that editor is **Vim**, which is hard to use if you don't know it. Tell Git to use **VS Code** instead:

```
git config --global core.editor "code --wait"
```

 **macOS:** first make the `code` command available. In VS Code, press `Cmd + Shift + P` and run **Shell Command: Install 'code' command in PATH**. Then open a new terminal.

Check it — you should see your name, your email, and the editor:

```
git config --global --list
```

2c — Get the course material

In the terminal, go to the folder where you want to keep the course (for example your *Documents* folder). Then **clone** the course repo, and create **your own branch**:

```
cd ~/Documents
git clone https://github.com/mkellnhofer/web-development-course.git
cd web-development-course
git switch -c my-work
```

`git switch -c my-work` creates a new branch called `my-work` and switches to it. All your own work goes there.

 **Never commit on `main`.** `main` holds the course material. If you commit on it, getting new material later fails.

Now open the course in VS Code: **File** → **Open Folder...** → `web-development-course`. The terminal in this window starts in the course folder — all following commands expect that. From here on, you can read this sheet in VS Code: open `exercises/week01/0/README.md` and press `Ctrl + Shift + V` (macOS: `Cmd + Shift + V`) for the preview.

Check it — the `*` marks the branch you're on:

```
git branch
```

2d — Your first commit

1. Open `exercises/week01/0/setup.txt`. Replace `_____` with the output of `git --version`. Save the file.
2. Ask Git what changed:

```
git status
```

Git lists `setup.txt` as **modified**.

3. **Stage** the change — this chooses what goes into the next commit — and **commit** it:

```
git add exercises/week01/0/setup.txt
git commit -m "Add my Git version"
```

4. Check it — your commit is on top:

```
git log --oneline
```

Task 3 — Node.js

3a — Install Node.js

Download **Node.js 24 (LTS)** from <https://nodejs.org> and install it. On Linux, follow the instructions for Linux on the same page.

 **Lab PC:** if the installer asks for an administrator password, tell me.

Close VS Code and open it again (see the note at the top). Check it — both must print a version number, and Node's must start with `v24` :

```
node --version
npm --version
```

3b — Install the dependencies

The `starter/` folder is a tiny web project. Its `package.json` says which packages it needs. `npm install` downloads them into the folder `node_modules/` :

```
cd exercises/week01/0/starter
npm install
```

3c — Serve the page

Start a small web server for this folder:

```
npm start
```

Open <http://localhost:3000> in Chrome. You should see **Hello Web Development!** Your computer is now a web server — Block 1 shows what happens between the browser and the server.

The server keeps running in this terminal. You stop it later with `Ctrl + C` .

3d — Change the page

1. Open `exercises/week01/0/starter/index.html` in VS Code.
2. Change `Hello Web Development!` to `Hello` and your name, e.g. `Hello Anna!` . Save the file.

3. **Reload** the page in Chrome (`F5`). You see your change.

3e — Commit your change

Open a **second terminal** with the **+** in the terminal panel. The server keeps running in the first one. The new terminal starts in the course folder.

```
git status
```

You should see exactly **one** changed file: `index.html` . The folder `node_modules/` isn't listed, because `starter/.gitignore` tells Git to ignore it.

```
git add exercises/week01/0/starter/index.html
git commit -m "Say hello with my name"
git log --oneline
```

Your two commits are on top.

Task 4 — Get Block 1 (live)

Wait until I publish Block 1 at the end of the session. Then get it, the same way you'll get new material every week:

```
git switch main
git pull
git switch my-work
git merge main
```

1. `git switch main` — go to the course material
2. `git pull` — download what's new
3. `git switch my-work` — back to your own branch
4. `git merge main` — bring the new material into your branch

Git opens a tab in VS Code with a message for the merge. **Close the tab** to finish the merge.

Check it — the folder `exercises/week01/1/` is there now, and the log shows the merge:

```
git log --oneline --graph
```

 **Short version:** on your own branch, `git pull --no-rebase origin main` does all four steps at once. `--no-rebase` tells Git to *merge* the new material into your branch.

Done when...

- `[] git --version`, `node --version` and `npm --version` all print a version

- [] `git branch` shows `* my-work`
- [] Your page shows **Hello** and your name at <http://localhost:3000>
- [] `git log --oneline` shows your two commits
- [] After task 4: the folder `exercises/week01/1/` is there

Stretch goals (optional)

- Open the **Source Control** view in VS Code (`Ctrl + Shift + G`). Find your commits, and look at the change you made in `index.html` .
- Run `git log --oneline --graph --all` . Find `main` and `my-work` — it's the diagram from the slides.
- Learn more Git: the free book **Pro Git**, chapters 1–3: <https://git-scm.com/book>

Something went wrong?

- `git` , `node` , `npm` or `code` is "not recognized" / "not found": close VS Code completely and open it again. Still not found? Ask.
- Windows, PowerShell says "running scripts is disabled on this system": run this once, then try again:

```
Set-ExecutionPolicy -Scope CurrentUser RemoteSigned
```

- `git commit` says "Please tell me who you are": you skipped task 2b.
- Port 3000 is in use: `npm start` picks another port. Use the address it prints.
- The merge opened Vim anyway: type `:wq` and press `Enter` . Then do task 2b again.